

Lincoln Going

Dr. Pohly

CSCI245

Project 1: Quicksort

My Experiment

In my experiment I tested the program quicksort. I wanted to find out how the runtime and number of comparisons changed if I sorted an identical array multiple times. Would the results change when I repeated the test on a bigger size?

My proposed hypothesis was that the number of comparisons and running time would be consistent for each identical test because the code should function exactly the same.

To test my hypothesis I created a program that called quicksort inside of a nested for loop. The outer for loop ran through different sized arrays. The inner tested an identical array several times (7). The inner loop printed variables storing the size, amount of comparisons, and running time for each sort. The outer loop printed variables storing the average running time, max running time, min running time, and the difference between the max and min running time.

My results confirmed that the number of comparisons was the same for each time the identical array was tested. For arrays below ten thousand the running time was almost always zero. For arrays above ten thousand, running time differed from the average by 1 millisecond. The difference between the min and max running time was rarely greater than 2. Since this was consistent, between array size ten thousand and array size forty-five thousand, I assumed that the running time would stay relatively close to the average regardless of the size. However, when an array of five hundred thousand was tested, the difference between the max and the min varied from the thirties to the two hundreds. This showed me that while the number of comparisons does not change when any array is tested many times, the run time of each test varies at increasing capacities in relation to the size of the array, likely due to the computer processor and outside factors using the RAM. This concludes my findings.

quickSort

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

size: 10:

comparisons: 51

running times: 0

Avg: 0

Min: 0

Max: 0

Difference: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

size: 50:

comparisons: 334

running times: 0

Avg: 0

Min: 0

Max: 0

Difference: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

size: 100:

comparisons: 879

running times: 0

Avg: 0

Min: 0

Max: 0

Difference: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

size: 150:

comparisons: 1423

running times: 0

Avg: 0

Min: 0

Max: 0

Difference: 0

size: 500:

comparisons: 6177

running times: 0

size: 500:

comparisons: 6177

running times: 0

size: 500:

comparisons: 6177

running times: 0

size: 500:

comparisons: 6177

running times: 1

size: 500:

comparisons: 6177

running times: 0

size: 500:

comparisons: 6177

running times: 0

size: 500:
comparisons: 6177
running times: 0
Avg: 0
Min: 0
Max: 1
Difference: 1

size: 10000:
comparisons: 617836
running times: 3

size: 10000:
comparisons: 617836
running times: 2

size: 10000:
comparisons: 617836
running times: 2

size: 10000:
comparisons: 617836
running times: 2

size: 10000:
comparisons: 617836
running times: 2

size: 10000:
comparisons: 617836
running times: 2

size: 10000:
comparisons: 617836
running times: 2
Avg: 2
Min: 2
Max: 3
Difference: 1

size: 15000:
comparisons: 1291369
running times: 4

size: 15000:
comparisons: 1291369
running times: 4

size: 15000:
comparisons: 1291369
running times: 4

size: 15000:
comparisons: 1291369
running times: 4

size: 15000:
comparisons: 1291369
running times: 4

size: 15000:
comparisons: 1291369

running times: 4

size: 15000:

comparisons: 1291369

running times: 4

Avg: 4

Min: 4

Max: 4

Difference: 0

size: 20000:

comparisons: 2224284

running times: 7

size: 20000:

comparisons: 2224284

running times: 7

size: 20000:

comparisons: 2224284

running times: 7

size: 20000:

comparisons: 2224284

running times: 7

size: 20000:

comparisons: 2224284

running times: 7

size: 20000:

comparisons: 2224284
running times: 7

size: 20000:
comparisons: 2224284
running times: 7
Avg: 7
Min: 7
Max: 7
Difference: 0

size: 25000:
comparisons: 3402959
running times: 10

size: 25000:
comparisons: 3402959
running times: 10

size: 25000:
comparisons: 3402959
running times: 11

size: 25000:
comparisons: 3402959
running times: 11

size: 25000:
comparisons: 3402959
running times: 10

size: 25000:
comparisons: 3402959
running times: 11

size: 25000:
comparisons: 3402959
running times: 11
Avg: 10
Min: 10
Max: 11
Difference: 1

size: 30000:
comparisons: 4833456
running times: 15

size: 30000:
comparisons: 4833456
running times: 15

size: 30000:
comparisons: 4833456
running times: 15

size: 30000:
comparisons: 4833456
running times: 14

size: 30000:
comparisons: 4833456
running times: 15

size: 30000:
comparisons: 4833456
running times: 14

size: 30000:
comparisons: 4833456
running times: 15
Avg: 14
Min: 14
Max: 15
Difference: 1

size: 35000:
comparisons: 6553018
running times: 19

size: 35000:
comparisons: 6553018
running times: 21

size: 35000:
comparisons: 6553018
running times: 19

size: 35000:
comparisons: 6553018
running times: 20

size: 35000:
comparisons: 6553018

running times: 20

size: 35000:

comparisons: 6553018

running times: 20

size: 35000:

comparisons: 6553018

running times: 19

Avg: 19

Min: 19

Max: 21

Difference: 2

size: 40000:

comparisons: 8469513

running times: 25

size: 40000:

comparisons: 8469513

running times: 26

size: 40000:

comparisons: 8469513

running times: 25

size: 40000:

comparisons: 8469513

running times: 26

size: 40000:

comparisons: 8469513
running times: 25

size: 40000:
comparisons: 8469513
running times: 25

size: 40000:
comparisons: 8469513
running times: 26
Avg: 25
Min: 25
Max: 26
Difference: 1

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32

size: 45000:
comparisons: 10688305
running times: 32
Avg: 32
Min: 32
Max: 32
Difference: 0

size: 500000:
comparisons: 1256069091
running times: 3605

size: 500000:
comparisons: 1256069091
running times: 3615

size: 500000:
comparisons: 1256069091
running times: 3619

size: 500000:
comparisons: 1256069091
running times: 3617

size: 500000:

comparisons: 1256069091

running times: 3625

size: 500000:

comparisons: 1256069091

running times: 3602

size: 500000:

comparisons: 1256069091

running times: 3594

Avg: 3611

Min: 3594

Max: 3625

Difference: 31